GOUTAM GUPTA

Full-Stack MERN Architect & Android App Developer

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PROFESSIONAL SUMMARY

Proactive and results-driven Full-Stack MERN Architect and Android Application Developer with a strong track record of engineering scalable web applications, low-latency microservices, and high-performance native Android solutions. Proficient across modern JavaScript ecosystems (React.js, Node.js, Express, MongoDB Atlas) and native mobile frameworks (Java, Kotlin, Android SDK). Experienced in building real-time multiplayer WebSocket engines, AI-powered systems (Gemini AI, TFLite), and secure enterprise authentication (JWT, Nodemailer). Committed to clean architectural patterns, robust API design, and intuitive user experiences.

TECHNICAL SKILLS

- **â¢ Frontend Engineering:** React.js, Vite, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive UI/UX Design
- **â¢ Backend & APIs:** Node.js, Express.js, RESTful APIs, JWT Authentication, Nodemailer, WebSocket.
- **â¢ Mobile & Edge AI:** Android Studio, Java, Kotlin, Android SDK, Material 3, MPAndroidChart, Firebase, TFLite, OpenCV
- **â¢ Database Management:** MongoDB Atlas,
- **â¢ Tools & Integrations:** Git, GitHub, VS Code, Postman, Stripe API, Gemini API

FEATURED ENGINEERING PROJECTS

School Management System | Full Stack Web Platform

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, Tailwind CSS, JWT, Nodemailer, Gemini AI [Live Demo Source Code]

- â¢ Architected enterprise-grade school management platform with role-based access control (Admin, Teachers, Students, Parents).
- â¢ Engineered secure JWT authentication, Bcrypt password hashing, and Nodemailer OTP email verification workflows.
- â¢ Integrated AI-powered fee notification & reminder engine utilizing Google Gemini API with automated student progress tracking modules.

Neo Chess - Real-Time Multiplayer Engine | Mobile & WebSocket Engine

Tech Stack: Android Studio, Java, JavaScript, Node.js, Express, WebSocket, HTML5, CSS3 [Download APK Source Code]

- â¢ Developed real-time multiplayer chess game with native Android wrapper and high-performance embedded web execution layer.
- â¢ Engineered Node.js WebSocket backend server achieving sub-30ms player matchmaking, live move synchronization, and turn validation.
- â¢ Released standalone Android APK distribution package with offline fallback modes and zero-lag board rendering.

Cow Business - AI Dairy Farm Calculator | Android Fintech & Analytics App

Tech Stack: Java, Android SDK, Material 3, MPAndroidChart, PdfDocument, SharedPreferences, XML [Download APK Source Code]

- â¢ Created specialized native Android agricultural finance application for yield forecasting and dairy farm profitability analysis.
- â¢ Implemented reducing-balance loan EMI calculation algorithms and interactive visual cashflow graphs using MPAndroidChart.
- â¢ Engineered automated PDF export generator (PdfDocument) enabling farmers to save and share comprehensive business dossiers.

Sci-Fi Scientific Calculator | Native Android Calculator Application

Tech Stack: Kotlin, Android SDK, XML Layouts, Material 3, Gradle, Git, BigDecimal Precision Engine [Download APK Source Code]

- â¢ Built advanced native Android calculator with cyberpunk interstellar neon UI theme, matrix solvers, and complex equations.
- â¢ Engineered custom Java/Kotlin BigDecimal mathematical precision core preventing floating-point inaccuracies during multi-step conversions.
- â¢ Implemented built-in physics/math formula cheatsheets and released optimized native APK on GitHub Releases.

Invitor (Event Booking) Website | Full Stack Event Management

Tech Stack: React.js, Vite, Tailwind CSS, RESTful APIs, Postman API Testing [Live Demo Source Code]

- â¢ Built full-stack Event Booking and Ticketing management platform featuring dynamic role-based dashboards.
- â¢ Designed responsive cyberpunk-tailored interface using Tailwind CSS and high-velocity Vite build pipeline.
- â¢ Verified and audited end-to-end REST API communication with structured Postman collections for seamless reliability.

Full Stack Food Delivery Website | Full Stack MERN Platform

Tech Stack: React.js, Node.js, Express, MongoDB Atlas, Stripe Payment Gateway, Tailwind CSS [Live Demo Source Code]

- â¢ Developed end-to-end online food ordering application with live telemetry order status tracking and responsive user carts.
- â¢ Integrated Stripe payment gateway for encrypted online financial transactions with webhook-driven order fulfillment.
- â¢ Architected scalable backend REST APIs using Express.js and MongoDB aggregation queries for food catalog filtering.

EDUCATION

Bachelor of Technology (B.Tech) - Information Technology (2023 - 2026)
Kalyani Government Engineering College (KGEC / MAKAUT) â Score: 6.61 CGPA

Diploma in Electronics & Telecommunication Engineering (ETCE) (2020 - 2023)
Purulia Polytechnic (W.B.S.C.T.E) â Score: 79.9% Aggregate

Higher Secondary Education (10+2) (2018 - 2020)
Sodepur Colliery High School (W.B.C.H.S.E) â Score: 84.4%

Secondary School Education (10th Standard) (2018)
Sodepur Colliery High School (W.B.B.S.E) â Score: 59.86%

HONORS, RANKINGS & CERTIFICATIONS

- â¢ WBJELET State Joint Entrance Examination: General Rank 170 | OBC Rank 26 (Top Tier State Placement).
- â¢ Robo Fight Competition: 3rd Position Trophy representing Purulia Polytechnic in tactical robotics warfare.
- â¢ Open-Source Community: Regularly mentor junior peers in Android SDK configuration, React frontend patterns, and REST API design.
- â¢ Domestic Data Entry Operator (DDEO): Certified in database operational maintenance and speed typing.

- â¢ **Spring Framework for Java Developers: Practical Guide** **March 17, 2025**
  Udemy | Credential ID: UC-a44fa5d7-a5b0-4895-8512-126670934dbc | Duration: 20.5 Hours
  Mastered Spring Core, Dependency Injection (DI), Spring Boot, Spring Data JPA, and RESTful microservices architecture.
- â¢ **Power BI From Scratch** **December 21, 2025**
  Udemy | Credential ID: UC-3308c41b-93cb-485a-9ee3-610090f8ef62 | Duration: 40 Hours
  Mastered enterprise data modeling, DAX query optimization, Power Query ETL pipelines, and interactive analytics reporting dashboards.
- â¢ **CSS, Bootstrap, JavaScript And PHP Stack Complete Course** **May 8, 2024**
  Udemy | Credential ID: UC-895a323e-a202-41d4-a07e-08610799bbc2 | Duration: 8 Hours
  Gained expertise in responsive web design, DOM manipulation, asynchronous JavaScript operations, and PHP server scripting.
- â¢ **Learn the Python Programming Language** **March 16, 2024**
  Udemy | Credential ID: UC-90269a29-9c64-4fc6-8759-66586b6dc73e | Duration: 2.5 Hours
  Core proficiency in Python syntax, standard data structures, object-oriented concepts, and algorithmic problem-solving.
- â¢ **Common Indicators for Cryptocurrencies Trading with Python** **March 15, 2024**
  Udemy | Credential ID: UC-604e0fb5-86a3-400f-bbe4-4d8991b7fb4d | Duration: 2 Hours
  Implemented technical financial indicators, quantitative market data processing, and automated algorithmic calculations using Python.